Study Tracker System

1. Academic Informartion

-Subject Code: PRP230807

-Subject Name: Programming In Python

-Topic: Study Tracker System

2. Team Details

-B040 (Arya Parab)

-B043 (Vrushti Patel)

-B055 (Jiya Shah)

3. Project Overview

Study Tracker is a Flask-based web application designed to help students manage their study activities efficiently. It provides tools for tracking tasks, goals, study sessions, and productivity using features like a Pomodoro timer, analytics charts, and a calendar view.

4. Objectives

- Develop a full-stack web application using Python and Flask
- Enable users to manage study tasks, goals, and subjects
- Track study sessions and analyze productivity
- Implement authentication and user profile management
- Improve time management using Pomodoro technique

5. Tech Stack

- Backend: Python, Flask
- Database: SQLite3
- Frontend: HTML5, CSS3, JavaScript
- Visualization: Chart.js
- Version Control: Git & GitHub

6. Key Features

- User Registration and Login (Authentication system)
- Dashboard with productivity insights
- Task Management (Add/Edit/Delete tasks)
- Goal Tracking system
- Subject-wise organization
- Notes management
- Pomodoro Timer for focused study
- Calendar view for study sessions
- Charts and analytics for performance tracking
- User Profile management

7. Project Structure

```
study-tracker/
|
 ├── main.py # Main Flask application
 ├── models.py # Database models
 ├── requirements.txt # Project dependencies
|
 ├── instance/
| └── study_tracker.db # SQLite database
|
 ├── routes/
|  ├── auth.py # Authentication routes
| └── main.py # Core application routes
|
 ├── templates/ # HTML templates
|  ├── base.html
|  ├── dashboard.html
|  ├── login.html
|  ├── register.html
|  ├── tasks.html
|  ├── goals.html
|  ├── notes.html
|  ├── calendar.html
|  ├── profile.html
| └── sessions.html
|
```

```
├── static/
│   ├── css/style.css
│   └── js/
│       ├── app.js
│       ├── charts.js
│       └── pomodoro.js
```

8. Database Overview

The application uses SQLite and includes the following main entities:

- Users – stores user account and profile data

- Tasks – manages daily or scheduled tasks

- Goals – long-term objectives

- Subjects – categorization of study topics

- Sessions – records study sessions

- Notes – stores user notes

9. Installation & Setup

Step 1: Clone the repository

```bash

git clone <your-repo-link>

cd study-tracker

Step 2: Install dependencies

pip install -r requirements.txt

Step 3: Run the application

python main.py

Step 4: Open in browser

http://127.0.0.1:5000
```

10. Testing Checklist

- Registration and login functionality works correctly
- Dashboard loads with correct data
- Tasks and goals can be managed

- Notes can be created and saved
- Pomodoro timer functions properly
- Charts display accurate analytics
- Calendar shows study sessions

11. Future Enhancements

- Email notifications
- Mobile application version
- AI-based study suggestions
- Cloud database integration (PostgreSQL/MySQL)
- Export reports (PDF/Excel)

12. Conclusion

The Study Tracker System is a practical full-stack web application that helps students improve productivity and manage their study routines efficiently. It demonstrates the integration of backend development with a responsive frontend and data visualization tools.